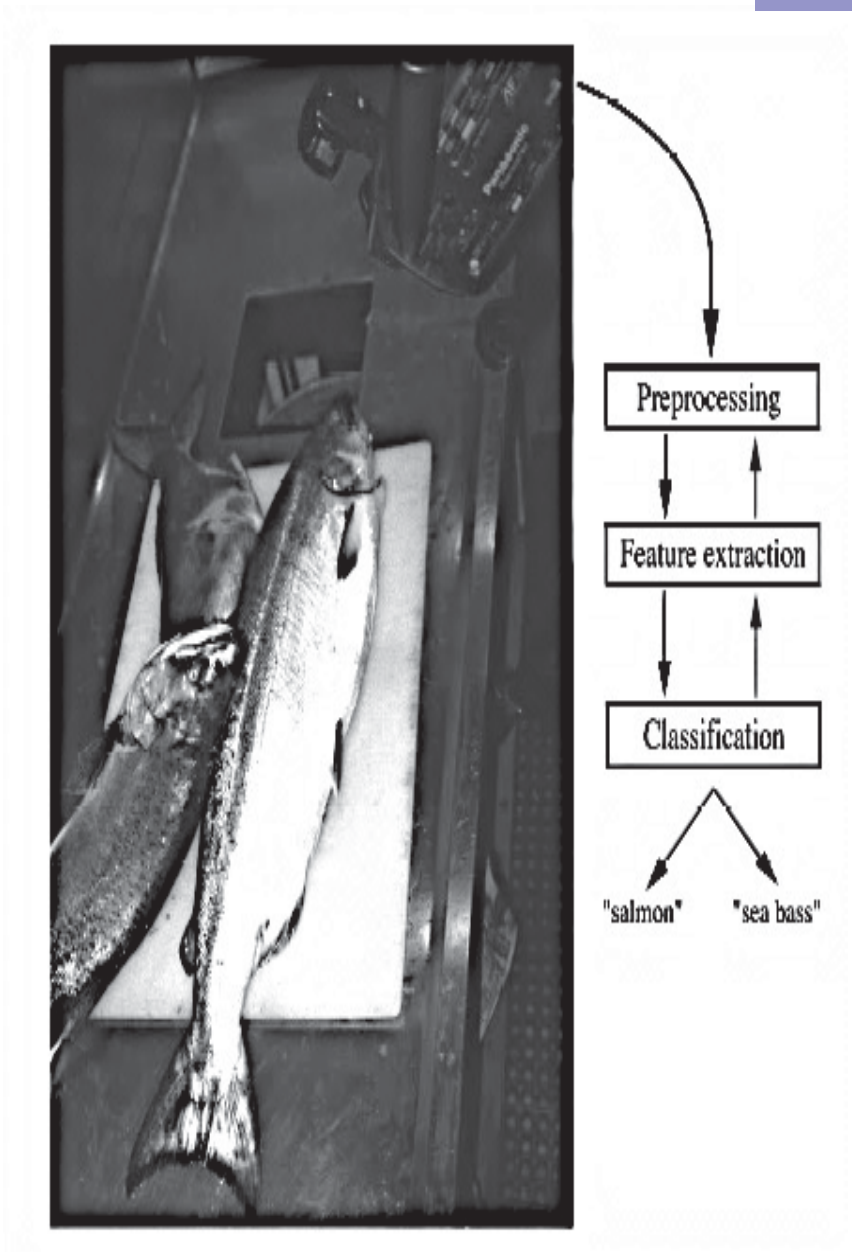
Tutorial 1:
practice & examples

An application example of machine learning:

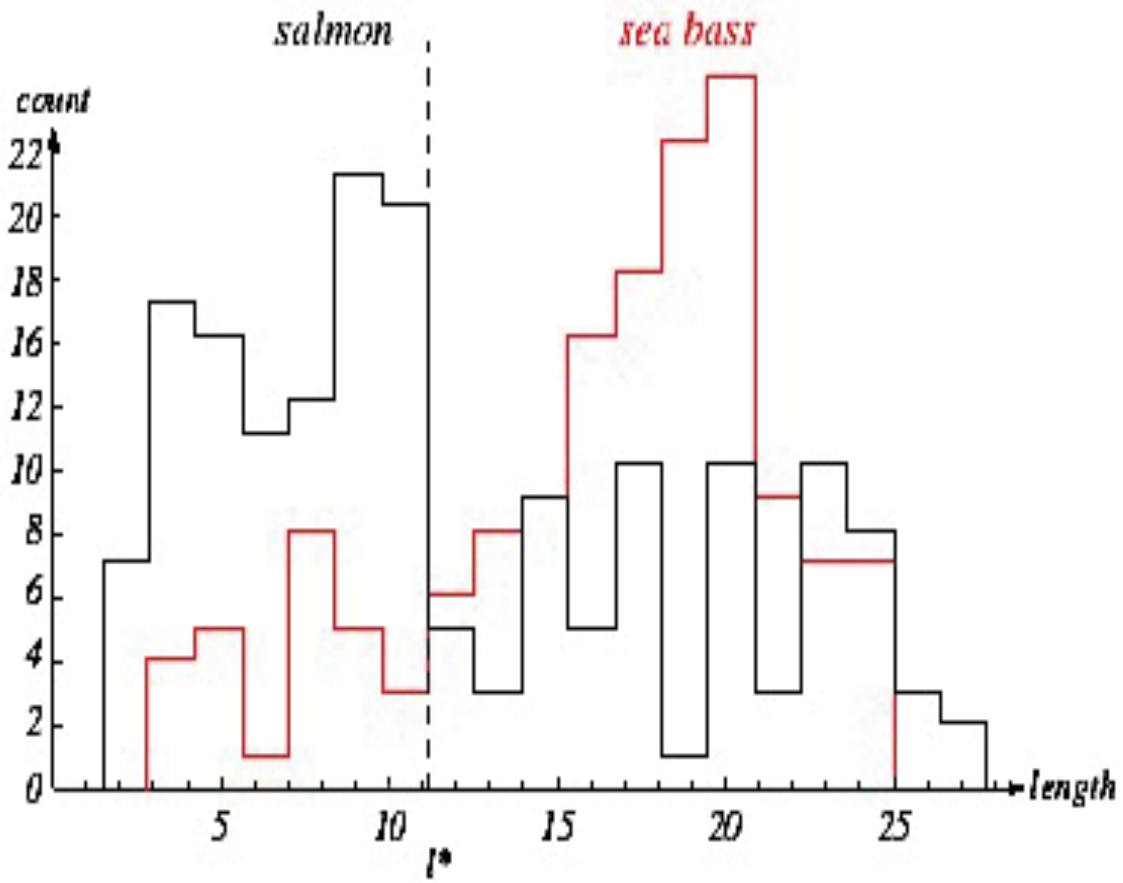
Sensing & Measurement → Data Pre-processing → Feature Extraction → Classification / Regression



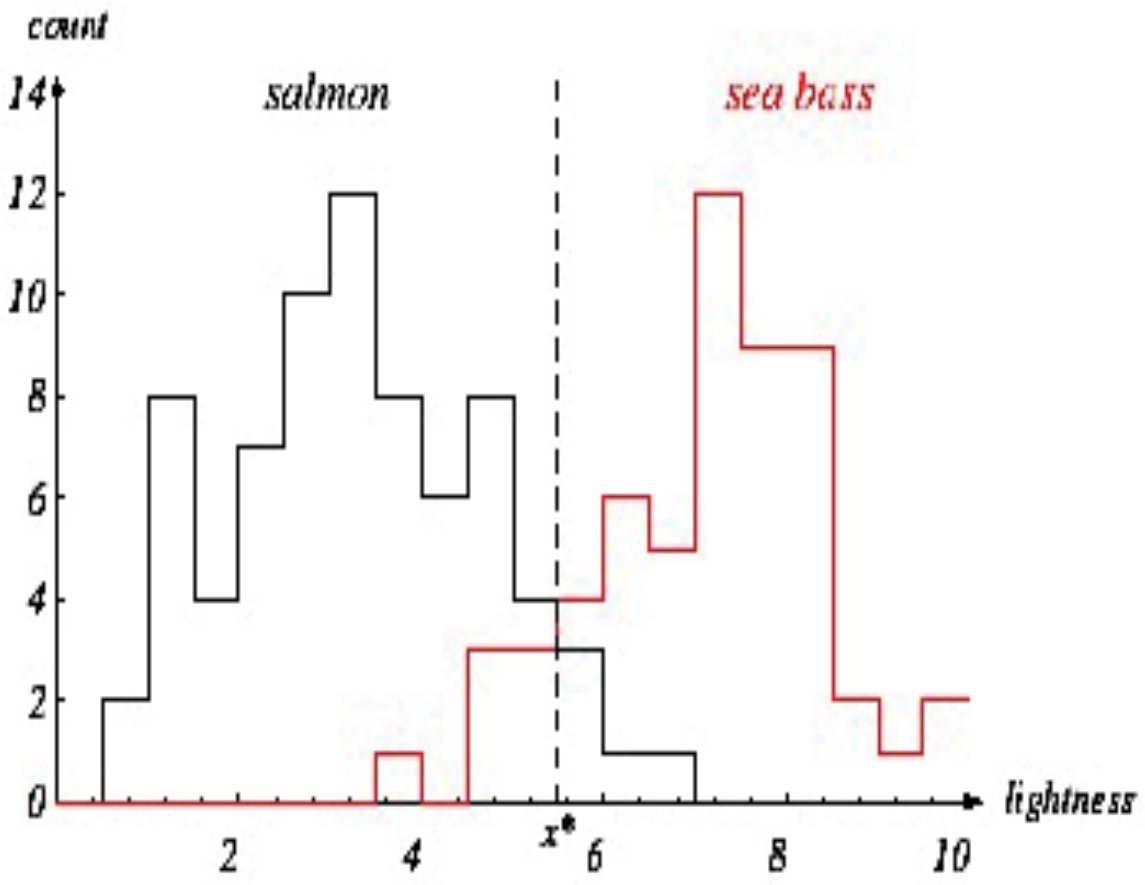
- Use a camera to calculate possible features for each fish:
  - length
  - colour / lightness
  - width
  - fin shape
  - mouth shape/angle

- These features may capture all the differences between salmon & sea bass.

Classification using **length**:



Classification using **lightness**:



Classification using **both features**:

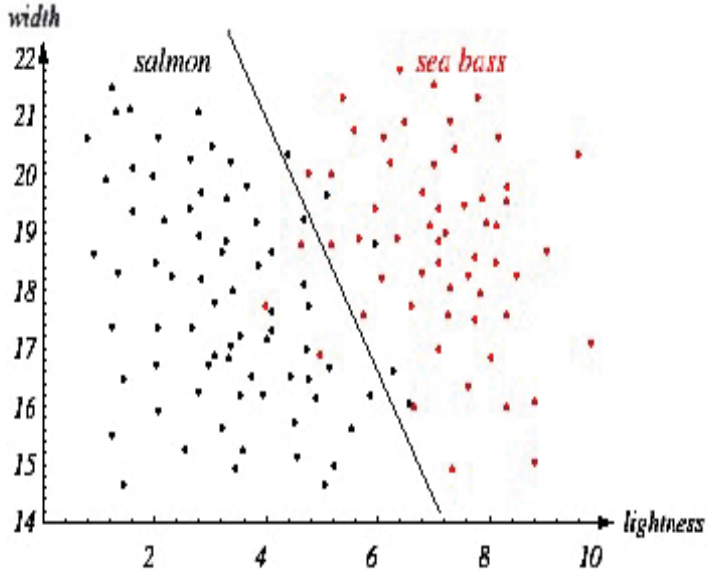
- Put both lightness and width in a single feature vector to represent each fish:

Fish → $x = [x_1, x_2]^T$

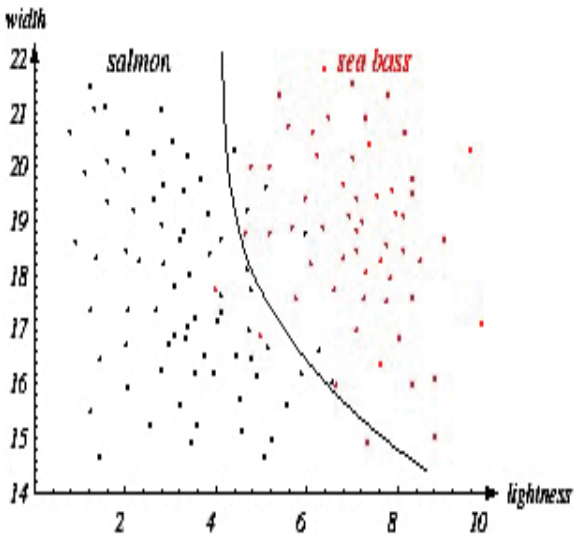
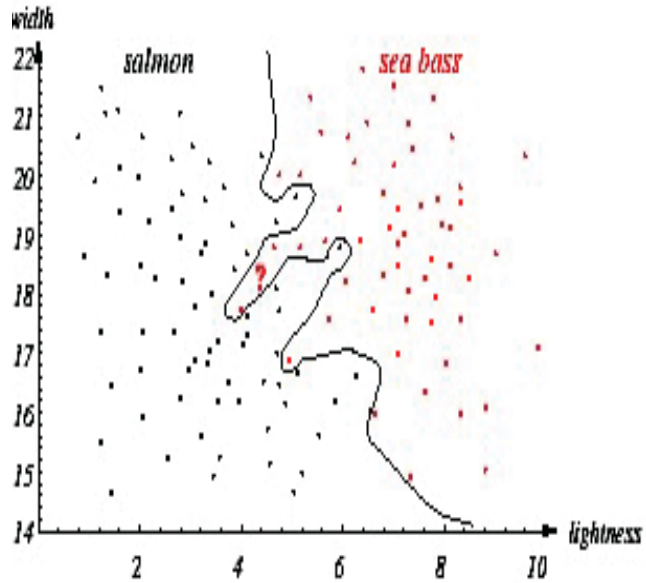
x_1 → lightness

x_2 → width

Possible decision boundaries:



Based on the training samples (set of fish **x** and its label), find the decision boundaries that generalise well to other unseen examples We measure the error with respect to a test set (unknown fish).

Underfitting: model not complex enough to accurately capture relationships between a dataset features and target variables.

Overfitting: model too closely fitting to a limited set of data points and does not generalise well to unseen data points.